

COVER PAGE for CBL STEM grants

Applicant name:	Jun Wang
Department:	Mechanical Engineering
School/College:	School of Engineering
Title of Proposal:	Parameterization Innovation: Can Human-Centered and Machine-Learned Representations Be Unified for Interpretable Design in Engineering?

Rank:

Assistant Professor

Award program:

Standard Research Award (up to \$15,000)

If you are asking for a Targeted Award, in which category are you applying? (Check one.)

N/A

If you are asking for a Standard Award, how would you describe your funding need? (Refer to guidelines for category definitions.)

Bridge funding

Are you asking for a course release? (If so, please note that you must have your chair's approval. Assistant Professors are not eligible for releases.)

No

Are you asking for a sabbatical award? (If so, please note that your sabbatical must be approved for the coming academic year to receive this award.)

No

Parameterization Innovation: Can Human-Centered and Machine-Learned Representations Be Unified for Interpretable Design in Engineering?

1 Project Description

Background Information: Finding effective representations for engineering design remains a persistent, fundamental, and open research problem [1]. It affects almost every primary design task, whether that be validation, optimization, manufacturing planning, uncertainty quantification, or understanding decision-making and preferences. The challenge of identifying useful representations arises from the need to consider both a design’s shape and function, as well as downstream engineering workflows, such as behavior (*i.e.*, validation) and embodiment (*i.e.*, manufacturing) (see Fig. 1). Algorithms have been created to jointly use or learn (in the machine-learning sense) such multifaceted representations. While machine-learned representations often perform better than human-specified ones on tasks—*e.g.*, design optimization, exploration, and innovation—as they prioritize internal efficiency (*e.g.*, generating thousands of CAD designs in a latent design space) that optimizes performance, they are often *not understandable or aligned with human (engineer) preferences* [2, 3]. How do engineers’ design heuristics (*e.g.*, avoid unsupported plates in sheet metal) shape machine-learned representations—and vice versa? Can one train machine learning (ML) systems that adjust representation formats based on engineer feedback or interaction patterns? What must one teach next-generation engineers so that they can address such multi-representational (*i.e.*, engineer-centered vs. machine-learned) interpretation and reasoning problems in the near future? This proposed project aims to address these and related questions by creating algorithms that learn from representations preferred by engineers and engineering workflows, and ultimately answer the fundamental *when* (a system shift from machine-learned to human-centered) and *how* (to find appropriate representations) questions.

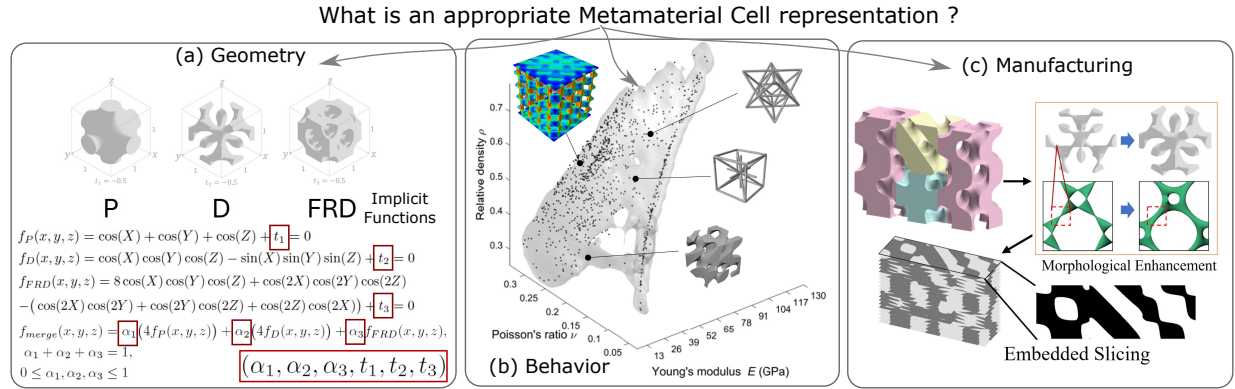


Figure 1: Example of Design Representations for metamaterial cells. The intricate metamaterial microstructures need multi-representational efforts to handle geometry, behavior, and manufacturing.

Research Objectives: If successful, such **unified engineer-centered and machine-learned representations**, while accelerating technological and economic competitiveness—*e.g.*, by improving optimization convergence by ten-fold, can (1) **match engineer domain concepts (Interpretability)**—*e.g.*, constraint-based CAD; (2) **focus on *what it does* rather than just *how it looks* (Reasoning)**—*e.g.*, design intent in parametric models; (3) **enable fast changes and *what-if* exploration (Usability)**—*e.g.*, interactive parametric sliders and constraint toggles; (4) **bridge multiple modalities for engineer fluency (Interoperability)**—*e.g.*, transitions across visual form (CAD), abstract math (latent spaces), and physical intuition (simulations).

Research Methods: I hypothesize that by **synergizing advances in (1) Neural Parameterization for geometry processing** (*e.g.*, Neural Implicit Representations, Signed Distance Functions) [4]—where neural networks are used to represent, manipulate, and generate complex shapes and visual data—with **(2) Approximation Theory for local refinement** (*e.g.*, T-Splines, LR-Splines, and PHT-Splines) [5]—which can provide a unified and more compact representation for both geometry and analysis for most real-world engineering designs—that algorithms can learn high-performing design representations of industrial-relevant problems with interpretable parameterizations for engineers. A student will be directly involved in the project through hands-on research on establishing the open-source testbeds, while contributing code, experiments, and publications.

Expected Outputs: The expected outcomes of this project are multifaceted and aim to contribute to the field with testbeds, publications, and grant proposals. **(1) Testbeds:** By way of analogy to ML, I expect to construct the equivalent of an ImageNet¹ or OpenAI Gym² **open-source software testbeds that compare the performance of design representations (whether machine-learned or human-centered).** **(2) Publications:** The methodologies and results under development are expected to deliver **two peer-reviewed journal articles** (e.g., *Computer-Aided Design*³, *Journal of Mechanical Design*⁴) while the preliminary results for future work are expected to be disseminated through **two conference presentations** such as *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC-CIE)* this August and *International Mechanical Engineering Congress & Exposition (IMECE)* this November. **(3) Grants:** I also expect to leverage the progress made through this project to submit a compelling NSF CAREER proposal this summer and pursue a DARPA Young Faculty Award (YFA) this November.

2 Project Timeline

Table 1 shows the task organization and the approximate timeline based on three categories: (1) a research plan for specific technical tasks, (2) a dissemination plan for paper drafting and conference presentations, and (3) a maintenance plan for sustaining the follow-up research. In addition, I will disseminate results via open-source code on GitHub. The proposed timeline is assumed to start from the summer quarter. For a one-year project, I will begin by constructing a single testbed to implement the proposed methods.

Table 1: Timeline of proposed activities.

	Tasks	Sum.	Fall	Win.	Spr.
Research	Select A Problem for Testbed Construction (e.g., Airfoil)	x			
	Implement Neural Parameterization Algorithms	x			
	Co-Design with Machine-Learned Parameters	x	x	x	
	Implement Locally Refined Approximation		x	x	
	Develop New ML Algorithms to Learn Approximation			x	x
	Compare with Baseline Algorithms				x
Dissem.	Abstracts & Posters for 2026 IDETC & IMECE	x			
	Manuscripts & Revisions for 2026 IDETC & IMECE	x	x		
	Extend Experiments & Results for Journals			x	x
	Prepare Full Manuscripts for Journals				x
Mainten.	Involve One Woman Student for Summer Research	x			
	Submit Full Proposal for 2026 NSF CAREER grant	x			
	Submit Project Summary for 2027 DARPA YFA		x		
	Prepare Full Proposal for DARPA YFA (if invited)			x	
	Integrate findings into Winter Machine Learning Course			x	
	Prepare A Proposal for NSF EDSE (if promising)				x

3 Broader Impact

Impact on Women in STEM: If awarded, I will use the CBL STEM award to intentionally create structured, sustained pathways for women to enter and advance in interdisciplinary research areas such as AI-driven engineering design and geometric modeling, where women remain underrepresented. Rather than assuming that inclusion will emerge organically from AI research, I will build inclusion directly into the project’s design, mentoring structures, and evaluation criteria. My goal is to ensure that women students not only participate in research but also gain early ownership, technical visibility, and leadership experience in mathematically rigorous domains. Specifically, I will integrate the funded research with inclusive educational and mentoring mechanisms, including: **(1) open-source datasets and tools with clearly defined, modular contribution pathways** that enable early and meaningful technical ownership; **(2) course-based research experiences** (e.g., *MECH 159: Engineering and Design with Applied Machine Learning*), in which I will intentionally scaffold women students into research roles through team-based projects, peer mentoring, and public dissemination of results (see Maintenance Activities in Table 1); and **(3) a paid, mentored summer research position reserved for one woman student**, whom I will directly supervise through structured weekly mentoring meetings, a milestone-driven research plan, and clearly defined scholarly outcomes such as an open-source contribution, a conference abstract, or a co-authored publication (see Budget Justification in Sec. 5). I will support women students as recognized contributors and emerging leaders, including as co-

¹ImageNet: <https://www.image-net.org/index.php>

²Gym: <https://gymnasium.farama.org/>

³Special Issue on AI for Geometric and Physical Modeling in Design and Manufacturing

⁴Special Issue on Generative Artificial Intelligence in Design

authors on scholarly outputs, maintainers of open-source resources, and near-peer mentors for subsequent cohorts. These experiences are designed to build research confidence, professional identity, and long-term career capital. In addition, by emphasizing interpretable, human-centered AI, I aim to align technical excellence with values of accessibility, ethics, and societal impact—broadening how success in STEM is defined and fostering research environments in which women can thrive as researchers, educators, and leaders.

4 Budget Breakdown

Table 2 shows the budget breakdown. The proposed project will be carried out by the PI and one student researcher (a woman student is preferred). I will work full-time on the research project, overseeing progress and developing compelling proposals for external grants. The student is expected to work on the project for 10 weeks during the summer.

Table 2: Budget breakdown for proposed project.

Category	Items	Costs
PI Summer Stipend	June 22, 2026 ~ August 28, 2026	\$10,000
Student Wages (One summer student)	June 22, 2025 ~ August 28, 2026 10 wks of summer x 23 hrs/wk x \$20/hrs	\$4,600
Benefits	Students SUM @ 8.5%	\$390

Total Requested: \$14,990

5 Budget Justification

Important Bridge Funding: This proposed research represents a continuous extension of my previously funded NSF project⁵. As illustrated in Fig. 1, I aim to generalize the data-driven design methodologies developed for metamaterial systems to a broader class of engineering design problems, such as airfoils, heat sinks, and waterline curves. This generalization is a critical next step for translating prior NSF investments into broadly applicable, domain-agnostic design frameworks. Specifically, the requested summer stipend will provide protected time to complete testbed construction and systematic data curation. These activities are not incremental extensions but enabling steps: a successful testbed will directly support the development of interpretable (in the machine-learning sense), transferable design representations across multiple engineering domains, substantially increasing the competitiveness of subsequent NSF and DoD proposals.

As I conclude my fourth year as an assistant professor, both my startup summer stipend and the remaining NSF funds will end in June 2026, creating a temporary but consequential funding gap with no summer salary support. Without bridge funding, this gap would significantly disrupt research continuity at a pivotal stage, risking loss of momentum precisely when prior results are mature enough to be generalized, validated, and positioned for competitive external proposals. I therefore view the CBL STEM award as essential bridge funding to sustain the lab, retain research focus, and complete high-leverage summer work that cannot be effectively conducted during the academic year.

In addition, the requested student summer wages will support the hiring of a woman student to participate fully in the summer research effort. This support will provide meaningful, mentored research experience in data-driven engineering design while directly contributing to core project deliverables, including dataset organization, validation studies, and preliminary modeling. In doing so, the award simultaneously advances technical objectives and broadens participation in STEM research.

Overall, this CBL STEM award would serve as a highly leveraged investment: it bridges a short-term funding gap, preserves the trajectory of NSF-supported research, enables concrete summer deliverables, and seeds compelling, externally competitive proposals.

References

- [1] Herbert A Simon. *The Sciences of the Artificial, reissue of the third edition with a new introduction by John Laird*. MIT press, 2019.
- [2] Yoshua Bengio, Aaron Courville, and Pascal Vincent. Representation learning: A review and new perspectives. *IEEE transactions on pattern analysis and machine intelligence*, 35(8):1798–1828, 2013.
- [3] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fournery, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. Guidelines for human-ai interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems*, pages 1–13, 2019.
- [4] Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. DeepSDF: Learning continuous signed distance functions for shape representation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 165–174, 2019.
- [5] Thomas W Sederberg, Jianmin Zheng, Almaz Bakenov, and Ahmad Nasri. T-splines and t-nurccs. *ACM transactions on graphics (TOG)*, 22(3):477–484, 2003.

⁵NSF-CDS&E-2245299, Collaborative Research: Data-Driven Inverse Design of Additively Manufacturable Aperiodic Architected Cellular Materials, awarded June 2023 (\$249,829; PI: Jun Wang, 100%).